

Opinion

The first thing I would investigate is the vector database and whether it is updating correctly. I would do this using an audit of the metadata in the vector database and checking whether it corresponds to the timestamps of the source documents.

The second thing I would check is the retriever, to see whether the similarity search is being performed correctly. First, I would probably inspect a recent sample of queries and their corresponding poor answers to check if there are any patterns or similarities between them. Then, I would select a small sample of the vector database (around 30-50 chunks) to do an evaluation using recall@k and MRR (Mean Reciprocal Rank). This checks for any inherent issues in the relevancy and ranking of the retriever when retrieving chunks.

The third thing I would check is the text embedding model, and whether there are any versioning issues due to provider-side updates. I would verify that the model version used during document ingestion time tallies with the version used at query time. I would also check the cache layer's TTL (Time To Live) to ensure that there isn't any stale data remaining in the embedding model that is producing outdated embedding.